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[This question paper contains 4 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 1687

F-8

Unique Paper Code : 2351601

Name of the Paper : Algebra V (Ring Theory & Linear Algebra - II)

Name of the Course : B.Sc. (H) Mathematics

Semester : VI

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All questions are compulsory.
3. Attempt any two parts from each question.

1. (a) If D is an integral domain, then prove that $D[x]$ is an integral domain.

(b) Let R be a commutative ring. Show that $R[x]$ has a subring isomorphic to R .

(c) Let $f(x) = x^3 + 2x + 4$ and $g(x) = 3x + 2$ in $\mathbb{Z}_5[x]$. Determine the quotient and remainder upon dividing $f(x)$ by $g(x)$.
(6,6,6)

P.T.O.

(c) Is $A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix}$ diagonalizable? $(6\frac{1}{2}, 6\frac{1}{2}, 6\frac{1}{2})$

5. (a) Prove that in an inner product space V , $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in V$: Verify the above inequality for $x = (2, 1 + i, i)$ and $y = (2 - i, 2, 1 + 2i) \in \mathbb{C}^3$.

- (b) Let V be an inner product space. Let S be orthogonal subset of V consisting of nonzero vectors. Prove that S is linearly independent.

- (c) Let W_1 and W_2 be subspaces of finite dimensional inner product space V . Prove that

$$(i) (W_1 + W_2)^\perp = W_1^\perp \cap W_2^\perp$$

$$(ii) (W_1 \cap W_2)^\perp = W_1^\perp + W_2^\perp \quad (6, 6, 6)$$

6. (a) For the data $\{(-1, 1), (0, 2), (1, 2), (2, 5)\}$, find a linear function of best fit using the least square approximation. Also, find the error E .

- (b) Let T be a linear operator on a finite dimensional complex inner product space V . Prove that T is normal $\Leftrightarrow$ there exists an orthonormal basis for V consisting of eigen vectors of T .

2. (a) State and prove Gauss's lemma.
 (b) Show that $1 - i$ is irreducible in $\mathbb{Z}[i]$.
 (c) In an Integral Domain, prove that every non-zero prime element is an irreducible element.

3. (a) For the function f defined on a vector space V , show that f is a linear functional where
 $V = \mathbb{R}^2$; $f(x, y) = (2x, 4y)$.
 (b) For the vector space V and its basis β , find the formula for vectors of the dual basis β^* .
 $V = \mathbb{R}^3$; $\beta = \{(1, 0, 1), (1, 2, 1), (0, 0, 1)\}$.
 (c) Define $f \in (\mathbb{R}^2)^*$ by $f(x, y) = 2x + y$ and $T \in \text{Hom}(\mathbb{R}^2, \mathbb{R}^2)$ by $T(x, y) = (3x + 2y, x)$. Compute $f(T(x, y))$.

4. (a) Let T be the linear operator on $P_2(\mathbb{R})$ defined by $T(f(x)) = f'(x)$. Let W be the T -invariant subspace generated by x^2 . Is W a T -invariant subspace?
 (b) Let T be a linear operator on a finite dimensional vector space V . Let $p(t)$ be the minimal polynomial of T . Show that λ is an eigen value of T if and only if λ is a root of $p(t)$. Hence, show that the characteristic polynomial and the minimal polynomial of T have the same roots.

- (c) Let V be a finite dimensional inner product space over F and $g: V \rightarrow F$ be a linear functional. Prove that there exists a unique vector $y \in V$ such that $g(x) = \langle x, y \rangle \quad \forall x \in V$.